

Manipal School of Information Sciences

Linux Operating System Lab Assignment

sed Stream Editor Lab

Authentication-Log and Web-Server-Log Analysis with sed

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sed Lab: Security Log Processing

Objective and setup

This lab uses the stream editor `sed` to filter, classify, mask and annotate two security logs: `auth.log` (SSH/sudo authentication events) and `access.log` (web-server requests). Every command below was executed and the output shown is the real output. The original log files are never modified except where an exercise explicitly asks for it (`-i` in Exercise 14, on a separate configuration file). Commands use GNU `sed`.

> **Line endings:** the supplied files use Windows (CRLF) line endings and have no newline at the end of the last line. A trailing carriage return would break end-of-line anchors such as `200$`, so the files were converted to Unix format first.

```
$ sed -i 's/\r$//' auth.log access.log      # CRLF -> LF (same as dos2unix)
[ -n "$(tail -c1 file)" ] && echo >> file      # add final newline if missing
wc -l auth.log access.log
  10 auth.log
  10 access.log
  20 total
```

```
$ cat -n auth.log
     1  Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
     2  Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
     3  Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
     4  Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
     5  Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
     6  Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
     7  Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
     8  Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
     9  Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
    10  Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

```
$ cat -n access.log
     1  192.168.1.10 - GET /index.html 200
     2  10.0.0.25 - GET /admin/login 401
     3  172.16.0.8 - GET /search?id=1'OR'1'='1 403
     4  192.168.1.15 - GET /products.html 200
     5  10.0.0.30 - GET /../../etc/passwd 403
     6  172.16.0.9 - POST /login 401
     7  192.168.1.20 - GET /contact.html 200
     8  10.0.0.25 - POST /admin/login 401
     9  172.16.0.8 - GET /script?input=<script> 403
    10  192.168.1.25 - GET /about.html 200
```

Part A: Authentication log (auth.log)

Exercise 1: Printing

1.1 Print the first log entry

```
$ sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
```

1.2 Print the fifth log entry

```
$ sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
```

1.3 Print entries from lines 2 to 6

```
$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
```

1.4 Print the last entry

```
$ sed -n '$p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

1.5 Print every second entry starting from line 1

```
$ sed -n '1~2p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

1.6 Print all entries containing Failed password

```
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

1.7 Print all successful logins (Accepted password)

```
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
```

1.8 Print from the first failed login to the authentication failure (range address)

```
$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

The range starts at the first line matching `Failed password` (line 2) and stops at the first later line matching `authentication failure` (line 7).

Exercise 2: Filtering security events

2.1 Failed login attempts

```
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

2.2 Successful login attempts

```
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
```

2.3 Entries associated with the root account

```
$ sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

2.4 Entries containing Invalid user

```
$ sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
```

2.5 All events related to 172.16.0.8 (dots escaped so they match literal dots)

```
$ sed -n '/172\.16\.0\.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
```

2.6 All events that do not contain Accepted password (`!` negates the address)

```
$ sed -n '/Accepted password/!p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
```

```
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

2.7 All events except those generated by sudo

```
$ sed -n '/sudo/!p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

2.8 Entries with either Failed password or Invalid user (multiple -e)

```
$ sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

Exercise 3: Removing irrelevant entries (original file untouched)

3.1 Delete successful login entries

```
$ sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

3.2 Delete entries containing Connection closed

```
$ sed '/Connection closed/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

3.3 Delete the first three entries

```
$ sed '1,3d' auth.log
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
```

```
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

3.4 Delete entries from lines 4 to 7

```
$ sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

3.5 Delete all entries associated with 192.168.1.15

```
$ sed '/192\.168\.1\.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

3.6 Delete every second entry starting from the first line

```
$ sed '1~2d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

3.7 Delete all events that do not contain Failed password

```
$ sed '/Failed password/!d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

```
$ wc -l auth.log   # original still has all 10 lines
10 auth.log
```

Exercise 4: Classifying security events

4.1 Accepted password to [SUCCESSFUL LOGIN]

```
$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
```

```
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

4.2 Failed password to [FAILED LOGIN]

```
$ sed 's/Failed password/[FAILED LOGIN]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

4.3 Invalid user to [INVALID USER]

```
$ sed 's/Invalid user/[INVALID USER]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: [INVALID USER] hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

4.4 authentication failure to [AUTHENTICATION FAILURE]

```
$ sed 's/authentication failure/[AUTHENTICATION FAILURE]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: [AUTHENTICATION FAILURE] for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

4.5 root to PRIVILEGED_USER

```
$ sed 's/root/PRIVILEGED_USER/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for PRIVILEGED_USER from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

4.6 Every occurrence of server to authentication-server (`g` flag)

```
$ sed 's/server/authentication-server/g' auth.log
Sep 10 09:10:12 authentication-server sshd[1201]: Accepted password for admin from
192.168.1.10
Sep 10 09:12:15 authentication-server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 authentication-server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 authentication-server sshd[1210]: Accepted password for student from
192.168.1.15
Sep 10 09:14:33 authentication-server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 authentication-server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 authentication-server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 authentication-server sshd[1230]: Accepted password for analyst from
192.168.1.20
Sep 10 09:18:05 authentication-server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 authentication-server sshd[1240]: Connection closed by 10.0.0.25
```

Exercise 5: Masking sensitive information

5.1 Replace 10.0.0.25 with X.X.X.X

```
$ sed 's/10\.0\.0\.25/X.X.X.X/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from X.X.X.X
Sep 10 09:12:20 server sshd[1206]: Failed password for root from X.X.X.X
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by X.X.X.X
```

5.2 Replace 172.16.0.8 with SUSPICIOUS_IP

```
$ sed 's/172\.16\.0\.8/SUSPICIOUS_IP/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from SUSPICIOUS_IP
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from SUSPICIOUS_IP
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

5.3 Replace username admin with USER_A

```
$ sed 's/\badmin\b/USER_A/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for USER_A from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for USER_A from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

5.4 Replace username student with USER_B

```
$ sed 's/\bstudent\b/USER_B/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for USER_B from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user USER_B
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

5.5 Replace all digits with X

```
$ sed 's/[0-9]/X/g' auth.log
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for admin from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for student from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Invalid user hacker from XXX.XX.X.X
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for admin from XXX.XX.X.X
Sep XX XX:XX:XX server sudo[XXXX]: authentication failure for user student
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for analyst from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Connection closed by XX.X.X.XX
```

5.6 Replace periods in IP addresses with hyphens

```
$ sed -E 's/([0-9]+)\.([0-9]+)\.([0-9]+)\.([0-9]+)/\1-\2-\3-\4/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192-168-1-10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10-0-0-25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10-0-0-25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192-168-1-15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172-16-0-8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172-16-0-8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192-168-1-20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10-0-0-30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10-0-0-25
```

5.7 Replace all IP addresses with [REDACTED_IP]

```
$ sed -E 's/[0-9]{1,3}(\.[0-9]{1,3}){3}/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
```

Part B: Web-server log (access.log)

Each record has the form `SOURCE_IP - METHOD URL STATUS`; the status code is the last field, so it is matched with the end-of-line anchor `$`.

Exercise 6: Printing web requests

6.1 Requests that returned 200

```
$ sed -n '/ 200$/p' access.log
192.168.1.10 - GET /index.html 200
192.168.1.15 - GET /products.html 200
192.168.1.20 - GET /contact.html 200
192.168.1.25 - GET /about.html 200
```

6.2 Requests that returned 401

```
$ sed -n '/ 401$/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
```

6.3 Requests that returned 403

```
$ sed -n '/ 403$/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
```

6.4 Requests to /admin/login (alternate delimiter `#` avoids escaping slashes)

```
$ sed -n '\#/admin/login#p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

6.5 All POST requests

```
$ sed -n '/ POST /p' access.log
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
```

6.6 Requests containing ../ (path traversal attempts)

```
$ sed -n '/\.\.\//p' access.log
10.0.0.30 - GET /../../etc/passwd 403
```

6.7 Requests containing <script> (cross-site scripting attempts)

```
$ sed -n '/<script>/p' access.log
172.16.0.8 - GET /script?input=<script> 403
```

6.8 Requests containing OR (SQL-injection indicator)

```
$ sed -n '/OR/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
```

6.9 All requests except successful ones

```
$ sed -n '/ 200$/!p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
```

```
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
```

6.10 All requests from 10.0.0.25

```
$ sed -n '/^10\.0\.0\.25 /p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

Exercise 7: Labelling suspicious web requests

7.1 200 to [ALLOWED]

```
$ sed 's/ 200$/ [ALLOWED]/' access.log
192.168.1.10 - GET /index.html [ALLOWED]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html [ALLOWED]
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html [ALLOWED]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html [ALLOWED]
```

7.2 401 to [UNAUTHORIZED]

```
$ sed 's/ 401$/ [UNAUTHORIZED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

7.3 403 to [BLOCKED]

```
$ sed 's/ 403$/ [BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200
```

7.4 /admin/login to /RESTRICTED_RESOURCE

```
$ sed 's#/admin/login#/RESTRICTED_RESOURCE#' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
```

```
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

7.5 GET to READ_REQUEST

```
$ sed 's/ GET / READ_REQUEST /' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
```

7.6 POST to SUBMIT_REQUEST

```
$ sed 's/ POST / SUBMIT_REQUEST /' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

Exercise 8: Grouping operations with { }

8.1 401 entries: relabel and print the modified line

```
$ sed -n '/ 401$/{s/ 401$/ [UNAUTHORIZED]/;p}' access.log
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.9 - POST /login [UNAUTHORIZED]
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
```

8.2 403 entries: relabel, redact the source IP, print

```
$ sed -n '/ 403$/{s/ 403$/ [BLOCKED]/;s/^[0-9.]*/[REDACTED_IP]/;p}' access.log
[REDACTED_IP] - GET /search?id=1'OR'1'='1 [BLOCKED]
[REDACTED_IP] - GET /../../etc/passwd [BLOCKED]
[REDACTED_IP] - GET /script?input=<script> [BLOCKED]
```

8.3 /admin/login requests: replace URL, redact IP, print

```
$ sed -n '\#/admin/login#{s#/admin/login#/PROTECTED_PAGE#;s/^[0-9.]*/[REDACTED_IP]/;p}'
access.log
[REDACTED_IP] - GET /PROTECTED_PAGE 401
[REDACTED_IP] - POST /PROTECTED_PAGE 401
```

Part C: Inserting, appending, changing and spacing

Exercise 9: Adding incident labels (i, a, c)

9.1 Insert a heading before the first line of auth.log

```
$ sed '1i SECURITY AUTHENTICATION LOG' auth.log
SECURITY AUTHENTICATION LOG
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

9.2 Append a message after the final entry

```
$ sed '$a END OF SECURITY LOG' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
END OF SECURITY LOG
```

9.3 Insert [WARNING] before every failed login

```
$ sed '/Failed password/i [WARNING]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
[WARNING]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
[WARNING]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
[WARNING]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
[WARNING]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

9.4 Append "Review this event" after every invalid-user entry

```
$ sed '/Invalid user/a Review this event' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Review this event
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
```

```
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

9.5 Insert [SUSPICIOUS WEB REQUEST] before every 403 entry

```
$ sed '/ 403$/i [SUSPICIOUS WEB REQUEST]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
[SUSPICIOUS WEB REQUEST]
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

9.6 Append [REQUIRES INVESTIGATION] after every 401 entry

```
$ sed '/ 401$/a [REQUIRES INVESTIGATION]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
[REQUIRES INVESTIGATION]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

9.7 Replace entries containing Connection closed with a label (c)

```
$ sed '/Connection closed/c [CONNECTION TERMINATED]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
[CONNECTION TERMINATED]
```

9.8 Replace all successful web requests with a label

```
$ sed '/ 200$/c [NORMAL WEB REQUEST]' access.log
[NORMAL WEB REQUEST]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
[NORMAL WEB REQUEST]
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
[NORMAL WEB REQUEST]
10.0.0.25 - POST /admin/login 401
```

```
172.16.0.8 - GET /script?input=<script> 403
[NORMAL WEB REQUEST]
```

Exercise 10: Separating entries with G

`G` appends the (empty) hold space to the pattern space, which prints as an extra blank line after the line.

10.1 Blank line after every authentication event

```
$ sed G auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15

Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

10.2 Blank line after every failed login

```
$ sed '/Failed password/G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

10.3 Blank line after every blocked web request

```
$ sed '/ 403$/G' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403

192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403

172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
```

```
192.168.1.25 - GET /about.html 200
```

10.4 Blank line after every second entry starting with the first

```
$ sed '1~2G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

Exercise 11: Writing matching events to files (w)

```
$ sed -n -e '/Failed password/w failed_logins.txt' -e '/Accepted password/w
successful_logins.txt' -e '/Invalid user/w invalid_users.txt' auth.log
```

```
$ sed -n -e '/ 401$/w unauthorized_requests.txt' -e '/ 403$/w blocked_requests.txt' access.log
```

```
$ sed -n '/172\.16\.0\.8/w suspicious_ip.txt' auth.log access.log
```

```
$
for f in failed_logins.txt successful_logins.txt invalid_users.txt unauthorized_requests.txt
blocked_requests.txt suspicious_ip.txt; do echo "--- $f"; cat $f; done
--- failed_logins.txt
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
--- successful_logins.txt
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
--- invalid_users.txt
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
--- unauthorized_requests.txt
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
--- blocked_requests.txt
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
--- suspicious_ip.txt
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
172.16.0.8 - GET /search?id=1'OR'1'='1 403
172.16.0.8 - GET /script?input=<script> 403
```

Exercise 12: Combining several operations with multiple -e

12.1 Label successful and failed logins

```
$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/' -e 's/Failed password/[FAILED LOGIN]/'
auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

12.2 Label 401 and 403 web requests

```
$ sed -e 's/ 401$/ [UNAUTHORIZED]/' -e 's/ 403$/ [BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200
```

12.3 Delete successful logins and print failed logins

```
$ sed -n -e '/Accepted password/d' -e '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

12.4 Replace root with [PRIVILEGED_USER] and mask all IP addresses

```
$ sed -E -e 's/root/[PRIVILEGED_USER]/g' -e 's/[0-9]{1,3}(\.[0-9]{1,3}){3}/[REDACTED_IP]/g'
auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
```

12.5 Replace GET and POST with READ_REQUEST and SUBMIT_REQUEST

```
$ sed -e 's/ GET / READ_REQUEST /' -e 's/ POST / SUBMIT_REQUEST /' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
```

```
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
```

Exercise 13: A sed script file (-f)

security_report.sed

```
1i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/
s/Invalid user/[INVALID USER ATTEMPT]/
s/Accepted password/[SUCCESSFUL LOGIN]/
s/[0-9]\{1,3\}\(\.[0-9]\{1,3\}\)\{3\}/[REDACTED_IP]/g
$a END OF INCIDENT REPORT
```

```
$ sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
```

```
$ sed -f security_report.sed auth.log > incident_report.txt
cat incident_report.txt
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
```

Exercise 14: Modifying a configuration file with sed -i

```
$ cat > security.conf <<'EOF'
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
EOF
cp security.conf security.conf.backup
cat security.conf
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
```

```
LogLevel INFO
GuestLogin enabled
```

All five changes are applied with one `sed -i.bak` call. Running `-i.bak` five separate times would overwrite `security.conf.bak` each time with an already-edited version, so only the last backup would be useful.

```
$ sed -i.bak -e 's/PermitRootLogin yes/PermitRootLogin no/' \
           -e 's/MaxAuthTries 6/MaxAuthTries 3/' \
           -e 's/FirewallStatus disabled/FirewallStatus enabled/' \
           -e 's/LogLevel INFO/LogLevel VERBOSE/' \
           -e 's/GuestLogin enabled/GuestLogin disabled/' security.conf
cat security.conf
echo '--- diff against backup ---'
diff security.conf.backup security.conf
PermitRootLogin no
PasswordAuthentication yes
MaxAuthTries 3
FirewallStatus enabled
LogLevel VERBOSE
GuestLogin disabled
--- diff against backup ---
1c1
< PermitRootLogin yes
---
> PermitRootLogin no
3,6c3,6
< MaxAuthTries 6
< FirewallStatus disabled
< LogLevel INFO
< GuestLogin enabled
---
> MaxAuthTries 3
> FirewallStatus enabled
> LogLevel VERBOSE
> GuestLogin disabled
```

`security.conf.bak` holds the original content, so the change can be reversed. This exercise edits a practice file only; it does not configure the real system.

Exercise 15: Bash script, security_log_analyser.sh

`security_log_analyser.sh`

```
#!/bin/bash
# security_log_analyser.sh - build a redacted alert report from an auth log
# Usage: ./security_log_analyser.sh <logfile>

REPORT="security_alert_report.txt"

# 1. argument supplied?
if [ $# -lt 1 ]; then
    echo "Usage: $0 <logfile>" >&2
    exit 1
fi
LOG="$1"

# 2. file exists (and is readable)?
if [ ! -f "$LOG" ]; then
    echo "Error: file '$LOG' not found." >&2
    exit 2
fi
if [ ! -r "$LOG" ]; then
    echo "Error: file '$LOG' is not readable." >&2
    exit 3
```

```
fi

TMP=$(mktemp)
trap 'rm -f "$TMP"' EXIT

# 3. extract failed-login and invalid-user attempts, redact IPs, flag with [ALERT]
sed -n -E '/Failed password|Invalid user/{
    s/[0-9]{1,3}(\.[0-9]{1,3}){3}/[REDACTED_IP]/g
    s/^/[ALERT] /
    p
}' "$LOG" > "$TMP"

FAILED=$(grep -c 'Failed password' "$LOG")
INVALID=$(grep -c 'Invalid user' "$LOG")

# 4. heading + summary + body + footer
{
    echo "SECURITY ALERT REPORT"
    echo "Source file : $LOG"
    echo "Failed logins : $FAILED"
    echo "Invalid-user attempts : $INVALID"
    echo "-------------------------------------"
    cat "$TMP"
    echo "-------------------------------------"
    echo "END OF REPORT"
} > "$REPORT"

# 5. display
cat "$REPORT"
```

Test run 1: normal use

```
$ chmod +x security_log_analyser.sh && ./security_log_analyser.sh auth.log
SECURITY ALERT REPORT
Source file : auth.log
Failed logins : 4
Invalid-user attempts : 1
-------------------------------------
[ALERT] Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
[ALERT] Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
[ALERT] Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
-------------------------------------
END OF REPORT
```

Test run 2: missing argument and missing file

```
$ ./security_log_analyser.sh; echo "exit status: $?"; ./security_log_analyser.sh nofile.log;
echo "exit status: $?"
Usage: ./security_log_analyser.sh <logfile>
exit status: 1
Error: file 'nofile.log' not found.
exit status: 2
```

Explanation. Argument and file checks come first so the script fails early with a clear message and a non-zero exit status. `sed -n -E '/A|B/{...}'` selects only the two event types; inside the block the IPs are redacted, then `[ALERT]` is added to the start of the line, then the line is printed. `grep -c` counts the events for the summary, and a `trap` removes the temporary file even if the script exits early.

Summary of sed commands used

Command	Purpose
`p`, `-n`	Print selected lines; `-n` suppresses automatic printing
`d`	Delete lines from the output (never from the file unless `-i` is used)
`s/old/new/g`	Substitute; `g` replaces every occurrence on a line
`i`, `a`, `c`	Insert before, append after, or change (replace) a line
`G`	Append the hold space, which adds a blank line
`w file`	Write matching lines to a file
`/re/,/re/`, `N~M`, `!`	Range, step (every Mth line) and negated addresses
`{ }`	Group several commands under one address
`-e`, `-f`, `-i[SUFFIX]`	Several expressions, a script file, in-place edit with optional backup